Answer Key - Physics - Undergraduate Level Assessment 16 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** D
   Options: A) 0.1 GeV/c^2; B) 0.2 GeV/c^2; C) 0.5 GeV/c^2; D) 1.0 GeV/c^2
   Note: Correct option: D - 1.0 GeV/c^2

2. **Answer:** C
   Options: A) electrons have many more degrees of freedom than atoms do; B) the electrons and the lattice are not in thermal equilibrium; C) the electrons form a degenerate Fermi gas; D) electrons in metals are highly relativistic
   Note: Correct option: C - the electrons form a degenerate Fermi gas

3. **Answer:** C
   Options: A) 1.2 x 10^9 m/s; B) 3.0 x 10^8 m/s; C) 1.5 x 10^8 m/s; D) 1.0 x 10^8 m/s
   Note: Correct option: C - 1.5 x 10^8 m/s

4. **Answer:** A
   Options: A) 4; B) 5; C) 6; D) 20
   Note: Correct option: A - 4

5. **Answer:** D
   Options: A) independent of M; B) proportional to $m^(1/2)$; C) linear in R; D) proportional to $R^(3/2)$
   Note: Correct option: D - proportional to $R^(3/2)$

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'The entropy of the system and its environment remains unchanged.'.

7. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '0.60 c'.

8. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'four hydrogen atoms and one helium atom'.

9. **Answer:** False
   Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'rate of change of the electric flux through S'.

10. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'an average of 10 times, with an rms deviation of about 3'.

Long Form Response

11. **Answer:** one-quarter as distant as they are today. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Hall coefficient. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key